

केस स्टडी/ CASE STUDY

OISD/ CS/ 2026-27/ P&E/12

Date: 08.09.2026

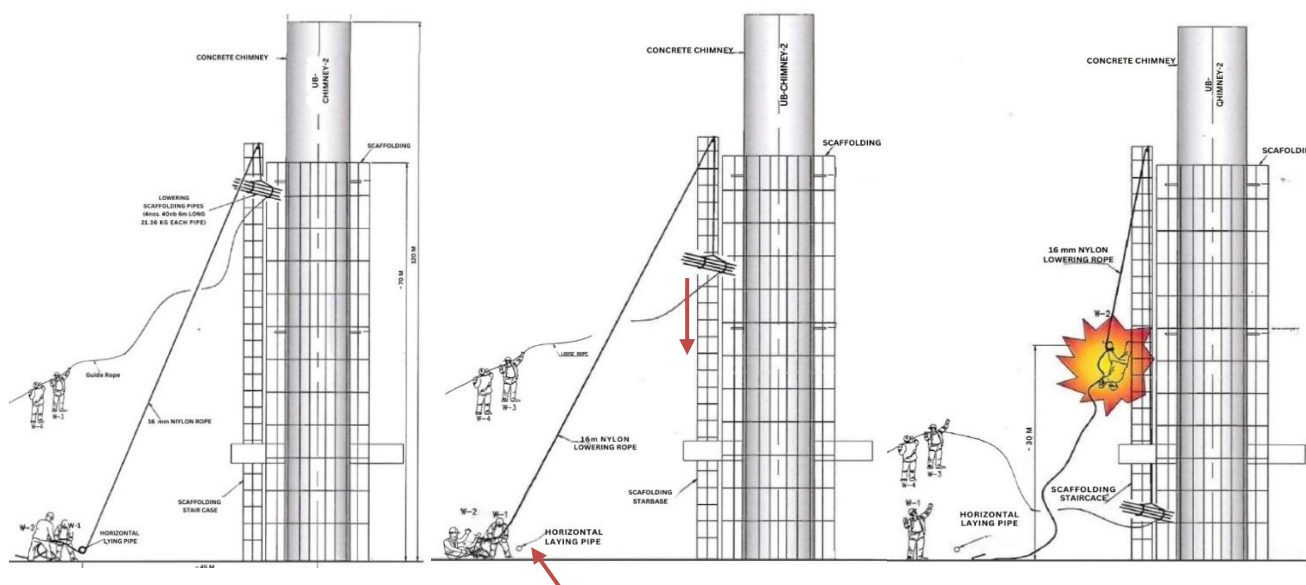
INTRODUCTION

Title: Fatal incident during scaffolding dismantling work
Location: Power Plant stack of a refinery
Loss/ Outcome: 1 Fatality

BRIEF OF INCIDENT

Dismantled scaffolding pipes were being manually lowered using a rope-and-pulley mechanism from the top of the scaffold (~70 m high) made for maintenance of stack (120 m high). The rope carrying the load was passed over a pulley installed at the top and then the rope was looped around a horizontal pipe on ground (for friction) before being controlled by two workers. One worker was standing in front and 2nd worker (injured person) standing behind him, while free end of the rope was laid haphazardly beside them.

During lowering the load, the front worker encountered a knot in rope, restricting the movement of rope looped around the horizontal pipe. While trying to release the knot, the looped roped came off the pipe and slipped from the hands of the workers leading to a free fall of the suspended load. The right leg of injured person somehow got entangled in the moving rope and he got lifted & pulled away towards the stack up to approx. 30 m height. Injured worker was rescued and shifted to nearby hospital where he succumbed to the injuries. Following graphics representation shows the sequence of events.



Two workers were handling the rope for lowering the scaffolding pipes and two workers were holding guide rope

Rope for lowering the load slipped off the horizontal lying pipe leading to freefall of the load

Free end of the rope got entangled in right leg of Injured Person and he got lifted

Provided for information purpose only. This information should be evaluated to determine if it is applicable in your operations, to avoid recurrence of such incidents.

OBSERVATIONS / LAPSES

1. The scaffolding pipes being lowered (allegedly 3 to 4 bundled each time) were 40 NB MS tubes, medium class 6 m in length, each weighing around 21.4 Kg. Total weight of the load including pipe and clamps was roughly 85-90 Kg. However, it could not be authenticated that not more than 4 pipes were lowered each time.
2. The rope used for lowering the dismantled scaffolding pipes, was passed over a pulley at top of scaffolding. On the ground a makeshift arrangement was used. The rope was looped around a 3" pipe inserted in 6" pipe and secured with wooden packings. The arrangement didn't have any standardised safety equipment like winch, free-fall arrester, stopper/ locking arrangement to prevent the rope from slipping off, etc.



3. Contractor's safety supervisor was authorized to receive work permit on behalf of industry's engineer in charge but was not given formal training on work permit as would have been to company employees issuing/receiving the work permit.
4. Issued work permit mentioned 20 work men were to be deployed for the job. But, as the actual number of workers was more, the same was modified to 20+20 by the contractor's safety supervisor without informing the same to permit issuer or requestor (engineer in charge of the job).
5. Because of additional work force requirements, contractor's site in charge called for 3 more workers working at another site without informing or taking permission from engineer-in-charge; Injured Person was one of them.
6. All workers were given training on Standard Maintenance Procedure (SMP) for scaffolding work, but the additional workers called for (including injured person), were not given similar training or explained with the hazards involved and required safety precautions.
7. The Job Safety Analysis (JSA) did not specify any upper limit on weight or number of pipes that shall be lowered at a time. Also, it did not envisage the possibility of free fall of material while lifting/ lowering and consequently did not provide control measures like fall arrester or mechanical braking system to avoid free fall, requirement for an approved rigging plan, method of communication etc.

The JSA was primarily based on the widely practiced qualitative judgment and individual perception which had gaps compared to quantitative methods of assessment.
8. Multiple unsafe conditions including one high potential near-miss involving fall of 75 Kg wet mixed mortar from a height of 100 meter were observed by engineer in charge of the job, but the same were not reported in safety portal and root cause analysis was not done.
9. Discrepancies in the attendance signatures of Toolbox Talk (TBT) records were observed raising concerns regarding the authenticity of TBT records.

Provided for information purpose only. This information should be evaluated to determine if it is applicable in your operations, to avoid recurrence of such incidents.

REASONS OF FAILURE / ROOT CAUSE

The cause of the incident was using of sub-standard practice of controlling the rope, used for lowering the load, through makeshift arrangement coupled with deployment of inadequately trained workers for the task of manual lowering a heavy suspended load.

The underlying reasons were inadequate risk control measures in Job Safety Analysis, not reporting and learning from previous near-miss incidents, inadequate training of contract supervisors and workers involved in critical job.

RECOMMENDATIONS

1. For any lifting/ lowering jobs, suitable engineered mechanisms (like manual or power operated Brake Winch with free fall arrest system) shall be explored and used.
2. The JSA for erection/ dismantling of high-rise scaffoldings should include requirement of task specific rigging plan for manual lifting/ lowering of heavy materials specifying load limits, method & equipment to be used, communication protocol, competency requirement, etc. Usage of quantitative methods for risk assessment in JSA may be explored.
3. The employees as well as contractual workforce should be sensitized and encouraged to report near misses, unsafe acts, and unsafe conditions so that such instances can be prevented.
4. For the same role, the competency and training requirements should be same for both employees as well as the contractor personnel. Hence, when contractor personnel are authorized to issue/ receive work permits, they should also be given the same initial as well as refresher training on work permit system that is given to company employees.
5. Control measures should be implemented to restrict deployment of contract workers from one site to another to ensure competence alignment, safety awareness with respect to the changed work environment/ scenario. All workers should be given specific training on SMP for critical jobs.
6. Toolbox talk covering job specific hazards and required safety precautions shall be judiciously conducted before commencement of the job. TBT shall be carried out with participation of all concerned workers and records maintained.